J. Dairy Sci. 106:4599–4607
https://doi.org/10.3168/jds.2022-22957
© 2023, The Authors. Published by Elsevier Inc. and Fass Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Effects of medium-chain fatty acids and tributyrin supplementation in milk replacers on growth performance, blood metabolites, and hormone concentrations in Holstein dairy calves

K. Murayama,[1,2] T. Fukui,[2] S. Kushibiki,[3] K. Sakamoto,[4] K. Inouchi,[1] and T. Sugino[2]*
[1]Dairy Technology Research Institute, National Federation of Dairy Co-operative Associations (Zen-Raku-Ren), Nishi-shirakawa, Fukushima, Japan 969-0223
[2]Research Center for Animal Science, Graduate School of Integrated Science for Life, Hiroshima University, Higashi-Hiroshima, Japan 739-8528
[3]Institute of Livestock and Grassland Science, NARO, Tsukuba, Ibaraki, Japan 305-0901
[4]YP Tech Co. Ltd., Chiyoda-ku, Tokyo, Japan 100-0004

ABSTRACT

This study aimed to evaluate the effects of triglycerides containing medium-chain fatty acids (MCT) and tributyrin (TB) supplementation in a milk replacer (MR) on growth performance, plasma metabolites, and hormone concentrations in dairy calves. Sixty-three Holstein heifer calves (body weight at 8 d of age, 41.1 ± 2.91 kg; mean ± SD) were randomly assigned to 1 of 4 experimental MR (28% crude protein and 18% fat): (1) containing 3.2% C8:0 and 2.8% C10:0 (in fat basis) without TB supplementation (CONT; n = 15), (2) containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation (MCT; n = 16), (3) containing 3.2% C8:0 and 2.8% C10:0 with 0.6% (dry matter basis) TB supplementation (CONT+TB; n = 16), (4) containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation (MCT+TB; n = 16). The MR were offered at 600 g/d (powder basis) from 8 to 14 d, up to 1,300 g/d from 15 to 21 d, 1,400 g/d from 22 to 49 d, down to 700 g/d from 50 to 56 d, 600 g/d from 57 to 63 d, and weaned at 64 d of age. All calves were fed calf starter, chopped hay, and water ad libitum. The data were analyzed using a 2-way ANOVA via the fit model procedure of JMP Pro 16 (SAS Institute Inc.). Medium-chain fatty acid supplementation did not affect the total dry matter intake. However, calves that were fed MCT had greater feed efficiency (gain/feed) before weaning (0.74 ± 0.098 vs. 0.71 ± 0.010 kg/kg) compared with non-MCT calves. The MCT calves also had a lower incidence of diarrhea compared with non-MCT calves during 23 to 49 d of age and the weaning period (50 to 63 d of age; 9.2% vs. 18.5% and 10.5% vs. 17.2%, respectively). Calves fed with TB had a greater total dry matter intake during postweaning (3,465 vs. 3,232 g/d). Calves fed TB also had greater body weight during the weaning (90.7 ± 0.97 vs. 87.9 ± 1.01 kg) and postweaning period (116.5 ± 1.47 vs. 112.1 ± 1.50 kg) compared with that of non-TB calves. The plasma metabolites and hormone concentrations were not affected by MCT or TB. These results suggest that MCT and TB supplementation in the MR may improve the growth performance and gut health of dairy calves.

Key words: medium-chain fatty acid, tributyrin, dairy calves

INTRODUCTION

In newborn dairy calves, ill health in their early life can have long-term negative effects on production. Additionally, the number of days that newborn calves have diarrhea during the first 4 mo of life has a significant negative impact on 305-d mature equivalent milk production and the production of actual milk, protein, and fat during the first lactation (Heinrichs and Heinrichs, 2011). The major role of the gastrointestinal epithelium is to protect the host from the negative effects of microorganisms, toxins, and chemicals in the intestinal lumen, and prevent unregulated movement of these compounds into the lymphatic and portal circulation (Gäbel et al., 2002; Steele et al., 2016). Thus, it is necessary to improve the growth and development of the gastrointestinal tract (**GIT**) in dairy calves.

Specific fatty acids and their triglycerides affect growth, internal secretions, and GIT development. Mills et al. (2010) reported that calves fed medium-chain triglyceride (**MCT**) rich in C8:0 had lower ADG compared with the control calves; however, feeding MCT rich in C12:0 did not affect ADG. Furthermore, calves fed MCT rich in C8:0 showed a greater decrease in the plasma glucose concentration after an insulin challenge. This result suggests that calves fed MCT rich in C8:0 promoted the uptake of glucose into

Received October 28, 2022.
Accepted January 15, 2023.
*Corresponding author: sugino@hiroshima-u.ac.jp

adipose tissue; however, there was no difference in the total empty body fat weight of calves fed MCT rich in C8:0 compared with that of control calves. Contrarily, supplementation of butyrate, coconut oil (rich in MCT), and flax oil (rich in linolenic acid) in milk replacer (**MR**) and calf starter increased the growth and immune and inflammatory responses by decreasing the change of rectal temperature and increasing IL-4 after vaccination of dairy calves (Hill et al., 2009, 2011; Quigley et al., 2019). Therefore, the effects of MCT supplementation and the synergistic effects of butyrate and MCT in dairy calves remain unclear.

Sodium butyrate supplementation in MR has positive effects on the growth performance (Górka et al., 2011) and GIT development of calves (Guilloteau et al., 2009; Górka et al., 2014). Elsabagh et al. (2017) reported that sodium butyrate stimulated glucagon-like peptide-2 (GLP-2) secretion in sheep. Glucagon-like peptide-2 is produced by enteroendocrine L cells of GIT, primarily from the distal intestine in response to nutrient intake (Burrin et al., 2003). It plays a role in intestinal growth stimulation in ruminants (Taylor-Edwards et al., 2011). Instead of sodium butyrate, tributyrin (**TB**) could be used as a butyrate source. Sodium butyrate is a short-chain fatty acid that contains 1 molecule of butyric acid, whereas TB is an ester composed of 3 molecules of butyric acid and glycerol. Furthermore, unlike sodium butyrate, TB does not have a butyrate-distinctive offensive odor and is a liquid at normal temperature. Therefore, TB could be a more practical supplement in MR compared with sodium butyrate. Araujo et al. (2016) reported that TB supplementation in MR did not affect the growth performance and metabolism of calves compared with sodium butyrate supplementation. Inabu et al. (2019) reported that feeding MR supplemented with TB increased the plasma concentration of GLP-2, but no difference was observed in growth performance.

However, there are only a few studies on TB supplementation in the MR of dairy calves, and its effects are unclear. We hypothesized that MCT and TB supplementation in MR would promote GIT development and growth performance. This study aimed to evaluate the effects of MCT and TB supplementation in MR on the growth performance, health, blood metabolites, and hormone concentrations of dairy calves.

MATERIALS AND METHODS

Animals and Housing

The procedures used in this study were performed in accordance with the principles and guidelines for animal use set out by Hiroshima University, and all experimental procedures were approved by the Animal Care and Use Committee of Hiroshima University (# E20–2). Sixty-three Holstein heifer calves (3 to 5 d of age; BW = 41.1 ± 2.91 kg, mean ± SD; Serum IgG at 8 d of age, 16.3 ± 1.34 mg/mL, mean ± SE) were transported from commercial dairies to the Dairy Technology Research Institute (Yabuki, Fukushima, Japan). The calves were born between July 3 and October 4, 2019. The calves were further blocked by birth date, BW, and farm origin and randomly assigned to 1 of the 4 treatments. The calves were raised outdoors in individual hutches (115 cm × 230 cm × 120 cm) that were bedded with a rubber mat on the bottom layer and sawdust on the top layer. When calves arrived at the research farm, they were checked for respiratory disease and diarrhea. They received 5 mL of Terramycin (Zoetis) and 0.5 mL of Duphafral Forte (Zoetis) via intramuscular injection and subcutaneous injection, respectively; calves also received 5mL of Ivelmec PO (Fujita Pharm) via percutaneous absorption. In addition to this, all the calves received 5mL of sulfonamides (Ektecin Liquid, Meiji Seika Pharma) daily for 3 consecutive days after their arrival. All the calves were fed 2 L of electrolyte solution on the day of their arrival. On the day after their arrival, they were fed MR (28% CP and 18% fat containing 3.2% C8:0 and 2.8% C10:0) without TB supplementation, up to 600 g/d (powder basis), using a bucket with a soft rubber nipple, twice daily at 0630 and 1630 h.

Feeding

All the calves were fed 1 of 4 experimental MR (166.7 g/L; Table 1): (1) containing 3.2% C8:0 and 2.8% C10:0 (in fat basis) without TB supplementation (**CONT**; n = 15), (2) containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation (**MCT**; n = 16), (3) containing 3.2% C8:0 and 2.8% C10:0 with 0.6% (DM basis) TB (BABY C4, SILO) supplementation (**CONT+TB**; n = 16), (4) containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation (**MCT+TB**; n = 16). Experimental MR (166.7 g/L) were offered at 600 g/d from 8 to 14 d, up to 1,300 g/d from 15 to 21 d, 1,400 g/d from 22 to 49 d, down to 700 g/d from 50 to 56 d, and 600 g/d from 57 to 63 d, and then weaned at 64 d of age. All the calves were fed a texturized calf starter and chopped Klein glass hay ad libitum from 8 d of age. The feeding time of solid feed was at 1000 h initially; however, when the calves started consuming more than 800 g/d (as fed) of starter, they were fed twice daily (equal amounts of starter at 1000 and 1500 h). Refused calf starter and hay were cleaned, and their amounts were recorded daily at 1000 h.

Table 1. Nutrient composition of milk replacer (MR), calf starter, and Klein grass hay

Item	MR[1]		Calf starter	Klein grass hay
	CONT and CONT+TB	MCT and MCT+TB		
DM, %	96.5	96.2	89.4	92.8
Nutrient component, % of DM				
CP	31.2	31.3	23.7	11.9
Fat	22.0	22.5	4.1	1.7
Ash	7.1	6.3	6.8	7.9
aNDFom	0.5	0.5	25.1	68.0
Starch	0.2	0.1	31.7	0.6
ME, Mcal/kg	4.87	4.93	3.41	2.36

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without tributyrin (TB) supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

Data and Sample Collection

The feed intake of individual calves was recorded daily. Fecal score (1 to 4 scale; 1 = normal fecal consistency to 4 = severe diarrhea) was recorded daily by 2 trained technicians. Disease incidence, if any, was recorded daily. Body weight, withers height, hip height, horizontal body length, heart girth, and hip width were measured at 0930 h at the start of the trial (8 d of age) and weekly thereafter until the end of the trial (91 d of age).

Blood samples were collected at 1000 h at 8 d of age and weekly thereafter, using vacutainers for the collection of plasma (Venoject II VP-H100K with heparin sodium; Terumo Corporation), serum (Insepack II SMD108CG with procoagulant and separating medium; Sekisui Medical), and whole blood (Venoject II VP-FH052K with sodium fluoride, heparin sodium, and EDTA-2Na, Terumo Corporation) from the jugular vein. Immediately after sample collection, aprotinin (500 kallikrein inhibitor units/mL of blood; Sigma-Aldrich Inc.) was added to the blood samples for plasma preparation. The blood samples were centrifuged at 1,800 × *g* at 4°C for 20 min, and both the plasma and serum were collected. Plasma and serum samples were stored at −80°C and −20°C, respectively, until analysis. Whole blood samples were stored at 4°C until analysis.

Sample Analysis

The calf starter and hay were sampled monthly and stored at room temperature. The samples were ground using a hammer mill (ST1; Retsch GmbH) with a 1-mm screen and analyzed by the Zen-Raku-Ren Analysis Center (Kamisu, Ibaraki, Japan) for DM, ash, CP, fat, and starch according to AOAC (1990), and for NDF according to AOAC International (2002).

Plasma blood samples were analyzed for IGF-1, GLP-2, and insulin concentrations. Plasma IGF-1 and GLP-2 concentrations were measured using the time-resolved fluoroimmunoassay technique previously described by Sugino et al. (2004). Plasma IGF-1 concentrations were measured using Eu-labeled human IGF-1 and polystyrene microtiter strips (Nalgene Nunc Int.) coated with goat anti-rabbit γ-globulin (Laarman et al., 2012). Plasma GLP-2 concentrations were measured using Eu-labeled human GLP-2 (Peptide Institute Inc.), polyclonal anti-rat GLP-2 (Yanaihara Institute Inc.), and polystyrene microtiter strips coated with goat-anti-rabbit γ-globulin (Elsabagh et al., 2017; Inabu et al., 2017). Plasma insulin concentrations were measured using the radioimmunoassay (Kushibiki et al., 2003). The intraassay coefficient of variation was 1.9%. Serum BHB and blood glucose concentrations were determined via enzymatic analysis using a clinical chemistry analyzer (Bio Majesty, JACK-BM, Jeol Ltd.).

Statistical Analysis

All the data were analyzed separately for the 4 periods differing in the amount of MR and primary nutrient sources: 8 to 22 d (low MR feeding during preweaning), 23 to 49 d (high MR feeding during preweaning), 50 to 63 d (weaning), 64 to 91 d of age (postweaning). All data were averaged by calf and period, and analyzed using JMP Pro 16 (SAS Institute Inc.), according to the following model:

$$Y_{ijk} = \mu + M_i + T_j + MT_{ij} + C_k + e_{ijk},$$

where Y_{ijk} is the dependent variable, μ is the overall mean, M_i is the fixed effect of MCT, T_j is the fixed effect of TB, MT_{ij} is the interaction between M_i and T_j, C_k is the random effect of the calf, and e_{ijk} is the residual. All values were expressed as the least squares means ± standard error of the mean. Significance was

Table 2. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on DMI (g/d) during 8 to 22 d and 23 to 49 d of age, weaning, and postweaning (LSM ± SEM)

	Treatment[1]					*P*-value		
Item, g/d	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)	SEM	MCT	TB	MCT × TB
8 to 22 d								
MR	763	759	759	767	5.4	0.95	0.78	0.25
Starter	41	21	36	33	12.2	0.27	0.71	0.39
Klein grass	3	2	3	3	0.8	0.52	0.23	0.55
Total DMI[2]	807	779	798	802	11.9	0.32	0.47	0.13
23 to 49 d								
MR	1,311	1,305	1,337	1,331	14.7	0.50	0.08	1.00
Starter	109	77	109	105	21.6	0.39	0.51	0.52
Klein grass	18	20	30	27	5.6	0.94	0.08	0.65
Total DMI[2]	1,438	1,398	1,476	1,459	27.9	0.30	0.08	0.69
Weaning								
MR	771	755	772	772	7.5	0.11	0.15	0.20
Starter	788	765	828	857	77.7	0.97	0.40	0.74
Klein grass	92	119	143	134	22.3	0.68	0.14	0.40
Total DMI[2]	1,649	1,638	1,744	1,760	78.7	0.97	0.17	0.87
Postweaning								
Starter	3,140	3,081	3,340	3,245	120.9	0.52	0.14	0.89
Klein grass	110	134	189	156	35.4	0.91	0.16	0.42
Total DMI[2]	3,247	3,217	3,528	3,402	113.8	0.50	0.05	0.68

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

[2]Total DMI was the sum of MR, calf starter, and hay.

declared at $P < 0.05$, and tendencies were declared at $0.05 \leq P < 0.10$.

RESULTS

One calf in the CONT group did not consume adequate MR, and another calf in the CONT+TB treatment group did not consume calf starter after weaning; these calves were excluded from the statistical analysis. Final treatment enrollment was as follows: CONT, n = 14; MCT, n = 16; CONT+TB, n = 15; MCT+TB, n = 16.

Intake and Growth

Medium-chain triglyceride supplementation in the MR did not affect DMI throughout the experimental period (Table 2). Tributyrin supplementation in the MR tended to increase the MR, hay, and total DMI at 23 to 49 d of age (1,334 vs. 1,308 g/d, $P = 0.08$; 29 vs. 19 g/d, $P = 0.08$; and 1,468 vs. 1,418 g/d, $P = 0.08$, respectively). Furthermore, TB calves had a greater total DMI than non-TB calves during the postweaning period (3,465 vs. 3,232 g/d; $P = 0.05$).

Medium-chain triglyceride supplementation in the MR did not affect BW, ADG, or skeletal development during 8 to 22 d of age (Table 3). During 23 to 49 d of age, still no differences were observed in BW, ADG, and skeletal development, but feed efficiency (gain/feed) was greater for the MCT calves compared with the non-MCT calves (0.74 vs. 0.71 kg of BW/kg of DM; $P = 0.03$). During the weaning period (Table 4), the MCT calves had a greater withers height gain compared with the non-MCT calves (0.29 vs. 0.23 cm/d; $P = 0.02$). No differences were observed in BW, ADG, skeletal development, or feed efficiency after weaning (Table 5).

Tributyrin supplementation in the MR did not affect BW and ADG, but horizontal body length and hip width gain were greater for the TB calves compared with the non-TB calves during 8 to 22 d of age (0.27 vs. 0.19, $P = 0.04$; and 0.09 vs. 0.06 cm/d, $P = 0.04$, respectively). During 23 to 49 d of age, the TB calves tended to have greater BW ($P = 0.09$) and heart girth gain ($P = 0.08$). The TB calves had greater BW than the non-TB calves during the weaning ($P = 0.05$; 90.7 vs. 87.9 kg) and postweaning periods (116.5 vs. 112.1 kg; $P = 0.04$). Effects of the interaction between MCT and TB were observed for feed efficiency at 8 to 22 d of age.

The MCT calves had lower fecal scores compared with the non-MCT calves (Table 6) at 23 to 49 d of age ($P = 0.03$; 1.7 vs. 1.8). Furthermore, the MCT calves had a lower incidence of diarrhea compared with the non-MCT calves during 23 to 49 d of age and the weaning period (9.2% vs. 18.5%, $P < 0.01$; and 10.5% vs. 17.2%, $P = 0.05$, respectively). Tributyrin supplementation in the MR did not affect the fecal

Table 3. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on growth performance during preweaning (LSM ± SEM)

	Treatment[1]					*P*-value		
Item	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)	SEM	MCT	TB	MCT × TB
8 to 22 d								
BW,[2] kg	46.3	45.7	46.8	46.7	0.74	0.65	0.29	0.77
ADG, kg/d	0.74	0.64	0.69	0.73	0.036	0.44	0.65	0.06
Withers height gain, cm/d	0.18	0.16	0.16	0.14	0.024	0.34	0.45	0.84
Hip height gain, cm/d	0.09	0.14	0.11	0.15	0.029	0.15	0.68	0.91
Heart girth gain, cm/d	0.36	0.37	0.31	0.35	0.041	0.63	0.42	0.70
Horizontal body length gain, cm/d	0.18	0.20	0.27	0.28	0.039	0.74	0.04	0.99
Hip width gain, cm/d	0.05	0.07	0.08	0.10	0.014	0.14	0.04	0.97
Feed efficiency[3]	0.84	0.71	0.74	0.83	0.053	0.67	0.83	0.05
23 to 49 d								
BW,[2] kg	70.9	69.6	71.7	72.1	0.97	0.67	0.09	0.39
ADG, kg/d	1.03	1.04	1.05	1.09	0.028	0.40	0.19	0.57
Withers height gain, cm/d	0.29	0.28	0.28	0.30	0.015	0.56	0.78	0.28
Hip height gain, cm/d	0.29	0.29	0.29	0.31	0.014	0.31	0.31	0.31
Heart girth gain, cm/d	0.47	0.46	0.48	0.52	0.021	0.60	0.08	0.34
Horizontal body length gain, cm/d	0.36	0.41	0.37	0.40	0.025	0.11	0.82	0.63
Hip width gain, cm/d	0.12	0.11	0.13	0.12	0.008	0.52	0.44	0.84
Feed efficiency[3]	0.71	0.74	0.71	0.75	0.014	0.03	0.88	0.66

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

[2]BW was the average of every weekly measurement.

[3]Feed efficiency was calculated as the ratio of ADG (kg/d) to total DMI (kg/d).

score or incidence of diarrhea. The calves used in the present study were generally healthy, and MCT and TB supplementation in the MR did not affect the number of days when veterinary treatment was needed (data not shown).

Metabolites and Hormones

Medium-chain triglyceride and TB supplementation in the MR did not affect blood glucose, plasma GLP-2, or IGF-1 concentrations throughout the experimental period (Table 7). Medium-chain triglyceride supplementation in the MR increased serum BHB concentrations (83.3 vs. 69.4 μmol/L; $P < 0.01$) during 23 to 49 d of age. During the weaning period, MCT supplementation in the MR tended to decrease the plasma insulin concentration ($P = 0.08$).

Tributyrin supplementation in the MR increased serum BHB concentrations (80.5 vs. 72.2 μmol/L; $P = 0.02$). During the weaning period, TB supplementation in the MR tended to increase the plasma insulin concentration ($P = 0.09$).

Table 4. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on growth performance during weaning (LSM ± SEM)

	Treatment[1]					*P*-value		
Item	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)	SEM	MCT	TB	MCT × TB
BW,[2] kg	88.1	87.7	89.9	91.6	1.40	0.66	0.05	0.43
ADG, kg/d	0.72	0.78	0.76	0.84	0.055	0.22	0.36	0.83
Withers height gain, cm/d	0.22	0.28	0.23	0.29	0.025	0.02	0.61	0.84
Hip height gain, cm/d	0.28	0.30	0.32	0.32	0.031	0.65	0.30	0.76
Heart girth gain, cm/d	0.34	0.32	0.37	0.38	0.037	0.88	0.30	0.69
Horizontal body length gain, cm/d	0.27	0.27	0.26	0.26	0.038	0.93	0.79	0.97
Hip width gain, cm/d	0.08	0.09	0.06	0.08	0.012	0.30	0.13	0.68
Feed efficiency[3]	0.42	0.46	0.44	0.47	0.025	0.15	0.48	0.82

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

[2]BW was the average of every weekly measurement.

[3]Feed efficiency was calculated as the ratio of ADG (kg/d) to total DMI (kg/d).

Table 5. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on growth performance during postweaning (LSM ± SEM)

Item	Treatment[1]				SEM	*P*-value		
	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)		MCT	TB	MCT × TB
BW,[2] kg	112.4	111.7	116.0	117.0	2.10	0.92	0.04	0.68
ADG, kg/d	1.25	1.21	1.28	1.28	0.054	0.77	0.39	0.67
Withers height gain, cm/d	0.19	0.21	0.21	0.21	0.018	0.70	0.45	0.61
Hip height gain, cm/d	0.20	0.21	0.21	0.21	0.020	0.70	0.88	0.71
Heart girth gain, cm/d	0.38	0.39	0.40	0.38	0.031	0.99	0.95	0.68
Horizontal body length gain, cm/d	0.30	0.29	0.33	0.31	0.025	0.54	0.34	0.94
Hip width gain, cm/d	0.11	0.10	0.11	0.12	0.009	0.97	0.31	0.62
Feed efficiency[3]	0.39	0.38	0.37	0.38	0.012	0.83	0.54	0.46

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

[2]BW was the average of every weekly measurement.

[3]Feed efficiency was calculated as the ratio of ADG (kg/d) to total DMI (kg/d).

DISCUSSION

Preweaning Period

Medium-chain triglyceride supplementation in the MR did not affect DMI during the preweaning period. However, growth performance was affected by MCT and TB supplementation in the MR. The MCT calves had greater feed efficiency during 23 to 49 d of age than the non-MCT calves. In a previous study, supplementation of milk with a blend of fatty acids (blended butyrate, coconut oil, and flax oil) increased feed efficiency during the nursery phase (Hill et al., 2011). Similarly, Quigley et al. (2019) demonstrated that calves that were fed MR with the same fatty acid blend as Hill et al. (2011) had greater feed efficiency, which is similar to our findings. However, Mills et al. (2010) demonstrated that calves that were fed MR with more than 30% of MCT of total fat had lower ADG compared with the control calves. In our study, the total content of C8:0 and C10:0 of total fat in the MR was 13.1%. Therefore, the calves in our study showed different growth performances compared with those in previous studies. Furthermore, in the present study, MCT supplementation in the MR reduced the fecal score and incidence of diarrhea during the preweaning period. Hill et al. (2011) reported that supplementation of MR with blended butyrate, coconut oil (rich in MCT), and flax oil (rich in linolenic acid) reduced abnormal fecal days and medical treatments. Additionally, C8:0 and C10:0 have been reported to

Table 6. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on fecal score and incidence of diarrhea during 8 to 22 d and 23 to 49 d of age, weaning, and postweaning (LSM ± SEM)

Item	Treatment[1]				SEM	*P*-value		
	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)		MCT	TB	MCT × TB
8 to 22 d								
Fecal score	2.1	2.0	2.0	2.0	0.10	0.41	0.46	0.78
Incidence of diarrhea,[2] %	32.0	24.6	25.0	25.0	4.72	0.44	0.49	0.44
23 to 49 d								
Fecal score	1.9	1.8	1.8	1.7	0.07	0.03	0.25	0.67
Incidence of diarrhea,[2] %	20.9	9.4	16.1	8.9	2.74	<0.01	0.34	0.43
Weaning								
Fecal score	1.8	1.7	1.8	1.6	0.09	0.07	0.61	0.77
Incidence of diarrhea,[2] %	17.9	11.6	16.5	9.4	3.34	0.05	0.60	0.89
Postweaning								
Fecal score	1.7	1.6	1.5	1.6	0.07	0.95	0.46	0.12
Incidence of diarrhea,[2] %	9.4	2.9	3.1	6.3	2.15	0.44	0.49	0.03

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

[2]Diarrhea: fecal scores 3 and 4.

Table 7. Effects of medium-chain triglyceride (MCT) and tributyrin (TB) supplementation in milk replacer (MR) on blood glucose, serum BHB, plasma insulin, and glucagon-like peptide-2 (GLP-2) levels during 8 to 22 d and 23 to 49 d of age, weaning, and postweaning (LSM ± SEM)

	Treatment[1]					*P*-value		
Item	CONT (n = 14)	MCT (n = 16)	CONT+TB (n = 15)	MCT+TB (n = 16)	SEM	MCT	TB	MCT × TB
8 to 22 d								
Glucose, mg/mL	102.7	98.1	99.6	98.8	2.79	0.33	0.67	0.52
BHB, μmol/L	52.7	55.8	54.6	58.6	2.57	0.17	0.36	0.86
Insulin, ng/mL	1.69	1.61	1.74	1.17	0.276	0.24	0.49	0.38
GLP-2, ng/mL	1.71	1.48	1.56	1.64	0.133	0.58	0.98	0.24
IGF-1, ng/mL	50.9	41.4	43.3	46.2	4.28	0.44	0.74	0.16
23 to 49 d								
Glucose, mg/mL	106.2	106.4	104.8	105.5	2.09	0.83	0.59	0.90
BHB, μmol/L	67.4	76.9	71.3	89.7	3.40	<0.01	0.02	0.20
Insulin, ng/mL	2.96	2.97	3.35	2.63	0.400	0.38	0.94	0.37
GLP-2, ng/mL	1.26	1.23	1.27	1.35	0.117	0.85	0.58	0.65
IGF-1, ng/mL	104.3	86.4	87.6	102.6	7.62	0.85	0.97	0.04
Weaning								
Glucose, mg/mL	99.7	98.2	98.3	100.6	2.13	0.86	0.79	0.37
BHB, μmol/L	178.0	170.9	163.1	178.8	11.34	0.70	0.76	0.32
Insulin, ng/mL	1.21	0.78	1.19	1.07	0.157	0.08	0.37	0.33
GLP-2, ng/mL	1.03	1.08	0.99	1.10	0.121	0.50	0.96	0.83
IGF-1, ng/mL	92.9	76.0	83.5	92.4	6.49	0.54	0.60	0.05
Postweaning								
Glucose, mg/mL	96.9	92.4	96.3	97.4	1.68	0.30	0.20	0.10
BHB, μmol/L	401.2	332.8	384.2	382.7	24.56	0.16	0.51	0.18
Insulin, ng/mL	0.84	0.82	1.02	1.05	0.114	0.96	0.09	0.84
GLP-2, ng/mL	1.04	1.03	1.00	0.98	0.120	0.87	0.73	0.96
IGF-1, ng/mL	123.9	108.8	120.7	121.7	8.90	0.43	0.59	0.37

[1]CONT was fed MR containing 3.2% C8:0 and 2.8% C10:0 (fat basis) without TB supplementation; MCT was fed MR containing 6.7% C8:0 and 6.4% C10:0 without TB supplementation; CONT+TB was fed MR containing 3.2% C8:0 and 2.8% C10:0 with 0.6% TB supplementation (DM basis); MCT+TB was fed MR containing 6.7% C8:0 and 6.4% C10:0 with 0.6% TB supplementation.

affect the intestinal microbiota and inhibit bacterial concentrations in in vitro experiments (Zentek et al., 2011). In the present study, the addition of MCT to MR may have affected the intestinal microbiota by increasing MR intake during 23 to 49 d of age and may have led to the reduction of diarrhea and increasing nutrient absorption. This is proposed to be the cause of the increased feed efficiency during the preweaning period. However, we did not evaluate the intestinal microbiota and nutrient digestibility. Further research is needed to evaluate the effect of dietary MCT supplementation on intestinal microbiota and gut health.

Tributyrin supplementation in MR increased BW during 23 to 49 d. Araujo et al. (2016) reported that 0.3% (DM basis) TB supplementation in MR did not affect BW in calves. Also, in our previous study (Inabu et al., 2019), 0.3% TB supplementation in MR did not affect growth performance in dairy calves. Calves in our study were fed 0.6% TB supplementation, double that of previous studies. Differences in feeding amounts of TB may have led to differences in responses to growth performance in calves. In addition, Górka et al. (2011) reported that the addition of sodium butyrate to MR positively affects fecal consistency in newborn calves. Furthermore, sodium butyrate supplementation in MR increases pancreatic secretions and nutrient digestibility (Guilloteau et al., 2009). In the current study, the development of GIT function may have improved growth performance. However, in the present study, TB supplementation in the MR did not affect the incidence of diarrhea, and we did not evaluate GIT development.

The impact of MCT and TB supplementation on blood metabolites and hormones was very limited. Only serum BHB concentrations were affected by MCT and TB supplementation during the preweaning period. Graulet et al. (2000) reported that the hepatic oxidation of MCT was greater than that of long-chain triglycerides, as observed from liver slices in preruminant calves. Blood BHB levels indicate rumen epithelial metabolic activity and development (Lane et al., 2000). Inabu et al. (2017) reported that the serum BHB concentration was positively correlated with the level of starter and hay intake in dairy calves. However, in the present study, solid feed intake during the preweaning period was low and did not differ among the treatments. In a previous study, stimulation of the forestomach development was observed in calves fed with MR containing butyrate (Górka et al., 2011; Kato et al., 2011). Supplementation of TB in the MR may indirectly stimulate forestomach development in the present study.

An interaction effect between MCT and TB was observed for feed efficiency during 8 to 22 d of age.

Single supplementation of MCT or TB decreased feed efficiency, although there was no significant difference among treatments. On the other hand, during 23 to 49 d of age, feed efficiency was increased by MCT supplementation to MR. Thus, the interaction effect between MCT and TB was inconsistent. Single supplementation of MCT or TB may have a negative effect on growth performance in the early life in dairy calves, but the effect was unclear in this study.

Weaning and Postweaning Period

Medium-chain triglyceride supplementation in the MR did not affect DMI during the weaning and postweaning periods. Compared with this, growth performance increased and the incidence of diarrhea decreased with the supplementation of MCT during the weaning period. These results suggest that the reduction in diarrhea could be because the addition of MCT may have affected the intestinal microbiota and increased the growth performance during the weaning period, similar to the preweaning period.

Total DMI during the postweaning period and BW during the weaning and postweaning periods were increased by TB supplementation. It is an interesting finding that BW after weaning increased even though calves were not given TB after weaning. In our knowledge, carryover effects of butyrate supplementation to MR were not investigated in dairy calves. Górka et al. (2011) reported that sodium butyrate supplementation stimulated rumen and small intestine development in calves compared with MR without sodium butyrate. Similarly, feeding fatty acid blends, including butyrate, increased the total-tract digestion of DM and nutrient components (Quigley et al., 2019). Furthermore, Inabu et al. (2019) reported that TB supplementation increased plasma GLP-2 concentrations in dairy calves, which might counterbalance the growth performance, despite decreased ME intake. These studies and our results suggest that TB supplementation in MR may promote nutrient digestibility by stimulating GIT development. As a result, the total DMI during the postweaning period and BW during the weaning and postweaning periods were increased in this study. However, we did not investigate intestinal growth and nutrient digestibility. In addition, MCT and TB supplementation in MR did not affect plasma GLP-2 concentration.

Glucagon-like peptide-2 has been a research target for ameliorating nutrient absorption and production efficiency in livestock (Burrin et al., 2003; Sigalet, 2012; Ipharraguerre et al., 2013). Elsabagh et al. (2017) reported that intraruminal injections of sodium butyrate increased plasma GLP-2 concentrations in sheep; Górka et al. (2009) reported that sodium butyrate supplementation in calf starters increased GLP-2 secretions in newborn calves. In our previous study, an increase in the plasma GLP-2 concentrations, due to TB supplementation in the MR, was observed in preweaning dairy calves (Inabu et al., 2019). However, in the current study, TB supplementation did not affect the plasma GLP-2 concentration. It has been shown that L cells secrete GLP-2 in a biphasic manner, in response to ingested nutrients (Connor et al., 2015): the initial release occurs within 30 min of nutrient intake via the indirect stimulation of subepithelial vagal afferent nerves, and then through subsequent efferent stimulation of the L cells by the enteric nervous system; a second peak in GLP-2 release occurs 1 to 2 h postprandially as a result of nutrients arriving in the distal intestine, as described (Xiao et al., 1999; Brubaker and Anini, 2003). In the current study, blood samples were collected 3 h after MR feeding; therefore, we may not have been able to detect a GLP-2 release peak at this time. Effects of MCT and TB supplementation in MR on GIT development and GLP-2 secretion need to be evaluated in future studies.

Interaction effects were observed for plasma IGF-1 concentration during weaning and incidence of diarrhea during postweaning. Both parameters were decreased by single supplementation of MCT or TB, although there was no significant difference among treatments. In addition, interaction effects between MCT and TB for metabolites and hormones and diarrhea were not observed during preweaning. Thus, the interaction effect between MCT and TB was inconsistent and unclear during postweaning, as for preweaning.

CONCLUSIONS

Both MCT and TB supplementation in MR increased BW and feed efficiency. Effects of MCT seemed to be decreased incidence of diarrhea during preweaning and weaning, whereas effects of TB seemed to be increased feed intake during postweaning. These results suggest that MCT and TB supplementation in MR may improve growth performance during the peri-weaning period in dairy calves.

ACKNOWLEDGMENTS

This work was supported by The National Federation of Dairy Co-operative Associations (ZEN-RAKUREN). The authors have not stated any conflicts of interest.

REFERENCES

AOAC. 1990. Official Methods of Analysis. 15th ed. Association of Official Analytical Chemists.

AOAC International. 2002. Official Methods of Analysis. 17th ed. AOAC International.

Araujo, G., M. Terré, A. Mereu, I. R. Ipharraguerre, and A. Bach. 2016. Effects of supplementing a milk replacer with sodium butyrate or tributyrin on performance and metabolism of Holstein calves. Anim. Prod. Sci. 56:1834–1841. https://doi.org/10.1071/AN14930.

Brubaker, P. L., and Y. Anini. 2003. Direct and indirect mechanisms regulating secretion of glucagon-like peptide-1 and glucagon-like peptide-2. Can. J. Physiol. Pharmacol. 81:1005–1012. https://doi.org/10.1139/y03-107.

Burrin, D. G., B. Stoll, and X. Guan. 2003. Glucagon-like peptide 2 function in domestic animals. Domest. Anim. Endocrinol. 24:103–122. https://doi.org/10.1016/S0739-7240(02)00210-2.

Connor, E. E., C. M. Evock-Clover, M. P. Walker, T. H. Elsasser, and S. Kahl. 2015. Comparative gut physiology symposium: Comparative physiology of glucagon-like peptide-2: Implications and applications for production and health of ruminants. J. Anim. Sci. 93:492–501. https://doi.org/10.2527/jas.2014-8577.

Elsabagh, M., Y. Inabu, T. Obitsu, and T. Sugino. 2017. Response of plasma glucagon-like peptide-2 to feeding pattern and intraruminal administration of volatile fatty acids in sheep. Domest. Anim. Endocrinol. 60:31–41. https://doi.org/10.1016/j.domaniend.2017.03.001.

Gäbel, G., J. R. Aschenbach, and F. Muller. 2002. Transfer of energy substrates across the ruminal epithelium: Implications and limitations. Anim. Health Res. Rev. 3:15–30. https://doi.org/10.1079/AHRR200237.

Górka, P., Z. M. Kowalski, P. Pietrzak, A. Kotunia, W. Jagusiak, and R. Zabielski. 2011. Is rumen development in newborn calves affected by different liquid feeds and small intestine development? J. Dairy Sci. 94:3002–3013. https://doi.org/10.3168/jds.2010-3499.

Gorka, P., Z. M. Kowalski, P. Pietrzak, A. Kotunia, R. Kiljanczyk, J. Flaga, J. J. Holst, P. Guilloteau, and R. Zabielski. 2009. Effect of sodium butyrate supplementation in milk replacer and starter diet on rumen development in calves. J. Physiol. Pharmacol. 60(Suppl 3):47–53.

Górka, P., P. Pietrzak, A. Kotunia, R. Zabielski, and Z. M. Kowalski. 2014. Effect of method of delivery of sodium butyrate on maturation of the small intestine in newborn calves. J. Dairy Sci. 97:1026–1035. https://doi.org/10.3168/jds.2013-7251.

Graulet, B., D. Gruffat-Mouty, D. Durand, and D. Bauchart. 2000. Effects of milk diets containing beef tallow or coconut oil on the fatty acid metabolism of liver slices from preruminant calves. Br. J. Nutr. 84:309–318. https://doi.org/10.1017/S0007114500001586.

Guilloteau, P., R. Zabielski, J. C. David, J. W. Blum, J. A. Morisset, M. Biernat, J. Wolinski, D. Laubitz, and Y. Hamon. 2009. Sodium-butyrate as a growth promoter in milk replacer formula for young calves. J. Dairy Sci. 92:1038–1049. https://doi.org/10.3168/jds.2008-1213.

Heinrichs, A. J., and B. S. Heinrichs. 2011. A prospective study of calf factors affecting first-lactation and lifetime milk production and age of cows when removed from the herd. J. Dairy Sci. 94:336–341. https://doi.org/10.3168/jds.2010-3170.

Hill, T. M., H. G. Bateman 2nd, J. M. Aldrich, and R. L. Schlotterbeck. 2009. Effects of changing the essential and functional fatty acid intake of dairy calves. J. Dairy Sci. 92:670–676. https://doi.org/10.3168/jds.2008-1368.

Hill, T. M., M. J. Vandehaar, L. M. Sordillo, D. R. Catherman, H. G. Bateman 2nd, and R. L. Schlotterbeck. 2011. Fatty acid intake alters growth and immunity in milk-fed calves. J. Dairy Sci. 94:3936–3948. https://doi.org/10.3168/jds.2010-3935.

Inabu, Y., K. Murayama, K. Inouchi, and T. Sugino. 2019. The effect of tributyrin supplementation to milk replacer on plasma glucagon-like peptide 2 concentrations in pre-weaning calves. Anim. Sci. J. 90:1185–1192. https://doi.org/10.1111/asj.13262.

Inabu, Y., A. Saegusa, K. Inouchi, S. Koike, M. Oba, and T. Sugino. 2017. Plasma concentrations of glucagon-like peptide 1 and 2 in calves fed calf starters containing lactose. J. Dairy Sci. 100:9361–9371. https://doi.org/10.3168/jds.2017-12910.

Ipharraguerre, I. R., G. Tedo, D. Menoyo, N. de Diego Cabero, J. J. Holst, M. Nofrarias, A. Mereu, and D. G. Burrin. 2013. Bile acids induce glucagon-like peptide 2 secretion with limited effects on intestinal adaptation in early weaned pigs. J. Nutr. 143:1899–1905. https://doi.org/10.3945/jn.113.177865.

Kato, S., K. Sato, H. Chida, S. G. Roh, S. Ohwada, S. Sato, P. Guilloteau, and K. Katoh. 2011. Effects of Na-butyrate supplementation in milk formula on plasma concentrations of GH and insulin, and on rumen papilla development in calves. J. Endocrinol. 211:241–248. https://doi.org/10.1530/JOE-11-0299.

Kushibiki, S., K. Hodate, H. Shingu, Y. Obara, E. Touno, M. Shinoda, and Y. Yokomizo. 2003. Metabolic and lactational responses during recombinant bovine tumor necrosis factor-α treatment in lactating cows. J. Dairy Sci. 86:819–827. https://doi.org/10.3168/jds.S0022-0302(03)73664-9.

Laarman, A. H., A. L. Ruiz-Sanchez, T. Sugino, L. L. Guan, and M. Oba. 2012. Effects of feeding a calf starter on molecular adaptations in the ruminal epithelium and liver of Holstein dairy calves. J. Dairy Sci. 95:2585–2594. https://doi.org/10.3168/jds.2011-4788.

Lane, M. A., R. L. Baldwin 6th, and B. W. Jesse. 2000. Sheep rumen metabolic development in response to age and dietary treatments. J. Anim. Sci. 78:1990–1996. https://doi.org/10.2527/2000.7871990x.

Mills, J. K., D. A. Ross, and M. E. Van Amburgh. 2010. The effects of feeding medium-chain triglycerides on the growth, insulin responsiveness, and body composition of Holstein calves from birth to 85 kg of body weight. J. Dairy Sci. 93:4262–4273. https://doi.org/10.3168/jds.2010-3142.

Quigley, J. D., T. M. Hill, L. E. Hulbert, T. S. Dennis, X. F. Suarez-Mena, and E. M. Bortoluzzi. 2019. Effects of fatty acids and calf starter form on intake, growth, digestion, and selected blood metabolites in male calves from 0 to 4 months of age. J. Dairy Sci. 102:8074–8091. https://doi.org/10.3168/jds.2019-16688.

Sigalet, D. L. 2012. Nonruminant Nutrition Symposium: The role of glucagon-like peptide-2 in controlling intestinal function in human infants: Regulator or bystander? J. Anim. Sci. 90:1224–1232. https://doi.org/10.2527/jas.2011-4704.

Steele, M. A., G. B. Penner, F. Chaucheyras-Durand, and L. L. Guan. 2016. Development and physiology of the rumen and the lower gut: Targets for improving gut health. J. Dairy Sci. 99:4955–4966. https://doi.org/10.3168/jds.2015-10351.

Sugino, T., Y. Hasegawa, Y. Kurose, M. Kojima, K. Kangawa, and Y. Terashima. 2004. Effects of ghrelin on food intake and neuroendocrine function in sheep. Anim. Reprod. Sci. 82–83:183–194. https://doi.org/10.1016/j.anireprosci.2004.05.001.

Taylor-Edwards, C. C., D. G. Burrin, J. J. Holst, K. R. McLeod, and D. L. Harmon. 2011. Glucagon-like peptide-2 (GLP-2) increases small intestinal blood flow and mucosal growth in ruminating calves. J. Dairy Sci. 94:888–898. https://doi.org/10.3168/jds.2010-3540.

Xiao, Q., R. P. Boushey, D. J. Drucker, and P. L. Brubaker. 1999. Secretion of the intestinotropic hormone glucagon-like peptide 2 is differentially regulated by nutrients in humans. Gastroenterology 117:99–105. https://doi.org/10.1016/S0016-5085(99)70555-X.

Zentek, J., S. Buchheit-Renko, F. Ferrara, W. Vahjen, A. G. Van Kessel, and R. Pieper. 2011. Nutritional and physiological role of medium-chain triglycerides and medium-chain fatty acids in piglets. Anim. Health Res. Rev. 12:83–93. https://doi.org/10.1017/S1466252311000089.

ORCIDS

K. Murayama https://orcid.org/0000-0003-3072-0693
T. Sugino https://orcid.org/0000-0001-5502-3451